

Question 1 / Vraag 1

Circle the correct answer / Omkring die korrekte antwoord

$$\frac{x+1}{-4} < 0 \Rightarrow$$

[1a] $x < -1$	[1b] $x > -1$	[1c] $x \in (-\infty, \infty)$	[1d] $x \in \{ \}$
[1e] None of these / Geen van hierdie			

Answer:

$$\frac{x+1}{-4} < 0 \Rightarrow x+1 > 0 \Rightarrow x > -1, \text{ so the answer is [1b]}$$

Question 2 / Vraag 2

Circle the correct answer / Omkring die korrekte antwoord

$$\frac{x+1}{1-x} \geq 0 \Rightarrow$$

[2a] $x \in (-\infty, -1) \cup (1, \infty)$	[2b] $x \in (-\infty, -1)$	[2c] $x \in (1, \infty)$	[2d] $x \in (-1, 1)$
[2e] None of these / Geen van hierdie			

Answer:

$$\frac{x+1}{1-x} \geq 0 \Rightarrow x+1 \geq 0 \text{ and } 1-x > 0 \text{ or } x+1 \leq 0 \text{ and } 1-x < 0$$

$$\Rightarrow x \geq -1 \text{ and } 1 > x \text{ or } x \leq -1 \text{ and } 1 < x \Rightarrow -1 \leq x < 1 \text{ or } x \in \{ \} \Rightarrow x \in [-1, 1), \text{ the correct answer is [2e]}$$

Question 3 / Vraag 3

Classify each function as a power function, root function, polynomial, exponential function, logarithmic function or rational function.

Klassifiseer elke funksie as 'n magsfunksie, wortelfunksie, polinoom, eksponensiaalfunksie, logaritmiëse funksie of rasionale funksie.

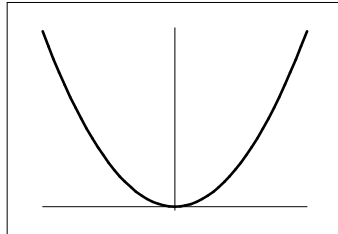
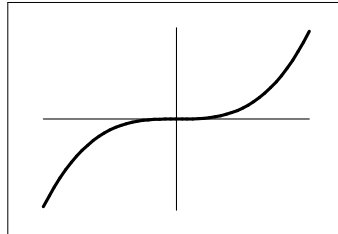
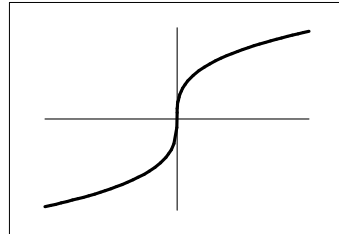
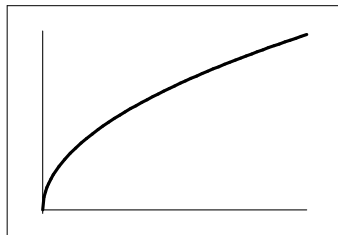
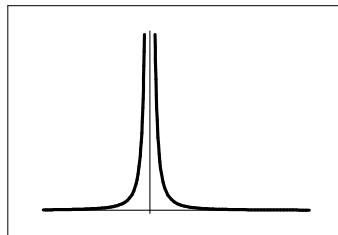
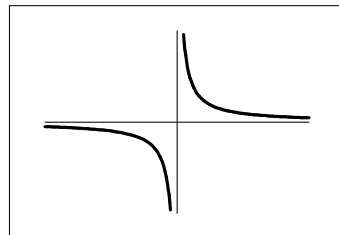
Answer: Sometimes more than one answer is correct

	Function / Funksie	Answer / Antwoord
(a)	$y = \sqrt{2}^x$	form $y = a^x$, exponential
(b)	$y = x^{\sqrt{2}}$	form $y = x^n$, power
(c)	$y = \sqrt{x}$	root or power
(d)	$f(t) = t^3$	power or polynomial
(e)	$g(u) = 3u^{10} - 4u^3 + 7$	polynomial
(f)	$y = \log_2 x$	logarithmic
(g)	$f(x) = \frac{2x^3}{1-x^2}$	rational

Question 4 / Vraag 4

Match the formula of the function to the graph of the function:

Pas die formule van die funksie byt die grafiek van die funksie:

Graphs / Grafieke**A****B****C****D****E****F**

	Formula / Formule	Answer / Antwoord
(a)	$y = x^7$	B
(b)	$y = x^{-7}$	F
(c)	$y = x^4$	A
(d)	$y = \frac{1}{x^4}$	E
(e)	$y = \sqrt[4]{x}$	D
(f)	$y = \sqrt[5]{x}$	C

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